

Safety inputs: 2 e-stops + 2 lidars, 24 V through PC817 – shield for Arduino UNO R4 WiFi / UNO Q

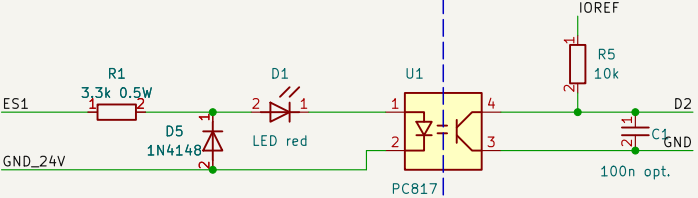
E-stop loops: 3.3 k → 6.3 mA (0.20 W at 28.8 V). Lidar inputs: 4.7 k → 4.4 mA (0.14 W). R1–R4 are 1/2 W (body 9 x 3.2 mm); pull-ups 1/4 W.

No current (button pressed, zone occupied, wire cut, 24 V lost) → pin HIGH = STOP. F1 protects the 24 V wires going out to the buttons.

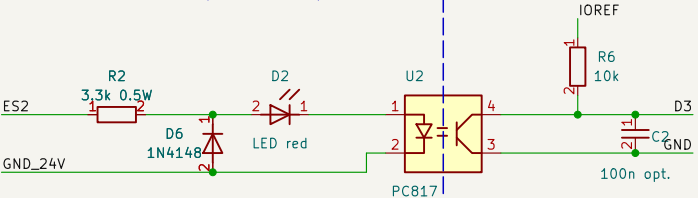
GND_24V and GND are NOT connected: this is the isolation barrier. Pull-ups go to IOREF (5 V on UNO R4 WiFi, 3.3 V on UNO Q), never to 5V.

C1–C4 are OPTIONAL: fitted, an opening shorter than about 1.5 ms is not seen by the Arduino, and every stop is seen about 1.6 ms later.

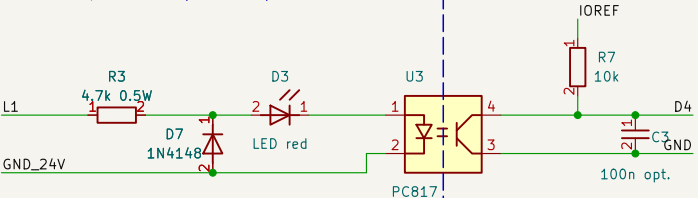
E-stop 1: button released (contact closed) → D2 LOW, LED lit



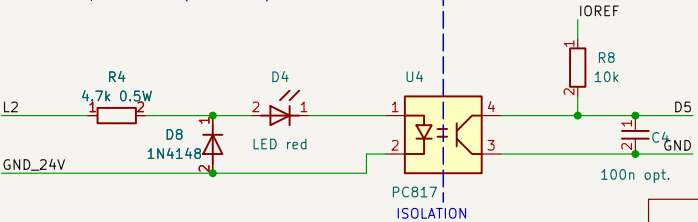
E-stop 2: button released (contact closed) → D3 LOW, LED lit



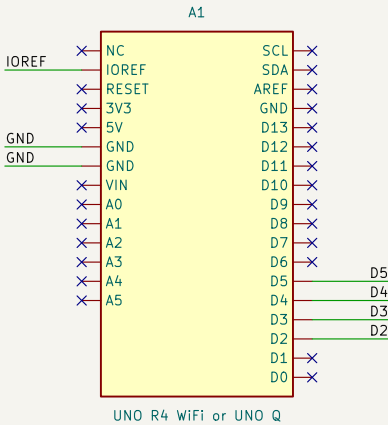
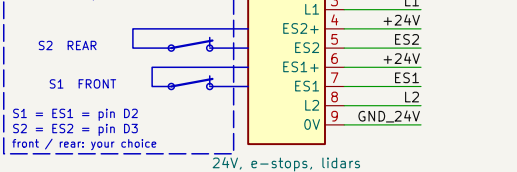
Lidar 1: output at 24 V (zone clear) → D4 LOW, LED lit



Lidar 2: output at 24 V (zone clear) → D5 LOW, LED lit



EXTERNAL – not on the board
Emergency-stop buttons:
1 NC contact, 2 terminals.
Pressed = open = STOP.



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